

Aditya Raj

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EDUCATION

IIST Shibpur (Indian Institute of Engineering Science and Technology)

B.Tech. Information Technology

Expected August 2027

Current CGPA: 8.34

SKILLS

Backend: Node.js, Express, fastAPI, GraphQL, PostgreSQL, MongoDB, Redis

Frontend: React, Next.js, TanStack Query, TailwindCSS

Infra/Tools: Docker, AWS, WebSockets, OAuth2, JWT, Git, VectorDB, Neo4j, RAG, AI Harness

Languages: C/C++, JavaScript, TypeScript, Python, SQL

EXPERIENCE

Software Engineer Intern

Remote

PhotoGPT | Certificate

2025 Aug – 2026 April (9 months)

- Led frontend orchestration for a **multi-model AI video generation** feature, coordinating payload construction, inference workflows, and response handling, driving **1.5+ hours** of user-generated video within the first week of launch.
- Architected and shipped a full **Social Media Publishing** system for an AI media platform, enabling users to publish AI-generated images/videos with captions, tags, and an admin moderation pipeline (pending/rejected states).
- Built SSR-powered creator profiles, follow system, and discovery feed with optimistic updates, optimized scalable media browsing and enabled seamless generation-to-sharing workflows using **React, Tanstack and Typescript**.

PROJECTS

MetaGraph-MCP - Active Metadata Platform | *Qdrant, Docker, Neo4j, Postgres, Typescript*

GitHub | Live Link

AUG 2026

- Architected a **dual-track metadata ingestion engine** that diffs live Postgres schema against a stored catalog and parses SQL via AST analysis to auto-derive table-level lineage - including foreign-key dependencies invisible to naive parsing - while an LLM-assisted agent generates business descriptions and column-level PII tags idempotently, at zero cost for unchanged tables.
- Built a **zero-polling, event-driven sync pipeline** on native Postgres LISTEN/NOTIFY (DDL + row-insert triggers) with debounced, requeue-safe processing, propagating live business-DB changes into a Neo4j lineage graph, Qdrant vector index, and governed catalog within seconds.
- Exposed the **governed catalog - lineage, schema, and semantic search** - as native MCP tools, enabling LLM agents to query PII-redacted, business-documented metadata directly, bridging traditional data governance with agentic AI workflows.

BubbleTea - Multi AI Agent Harness | *Typescript, MCP, Harness, CLI*

GitHub | Live Link

AUG 2026

- **Architected a multi-provider AI agent runtime** in TypeScript/Node.js - unified Gemini (native function-calling) and OpenRouter (hand-parsed SSE streaming, fragmented tool-call reassembly) behind one interface, driving a plan-act-observe tool-calling loop with a live-streaming Ink/React terminal UI.
- **Implemented parallel multi-agent orchestration and extensibility:** isolated sub-agents with scoped tool access running concurrently via an event-driven background task manager, plus a pluggable tool registry unifying built-in tools, MCP (Model Context Protocol) servers, and on-demand skills.
- **Engineered production-grade reliability and governance:** deterministic pre/post tool-call guardrail hooks, an append-only auditable session store with non-destructive context compaction, and a rule-based + LLM-judge evaluation loop with automatic bounded-retry repair.
- Built a focused **test suite** for an agentic CLI's core mechanics (tool-call protocol, safety guardrails, MCP tool discovery).

ACHIEVEMENTS

WebGame Competition | 1st place for engineering a real-time multiplayer game with WebSocket-driven state sync.

Data Structures & Algorithms | Solved 700+ problems on LeetCode and Codeforces.

Open Source | Contributed to open-source projects during GSSoC'23 and Hacktoberfest'24.